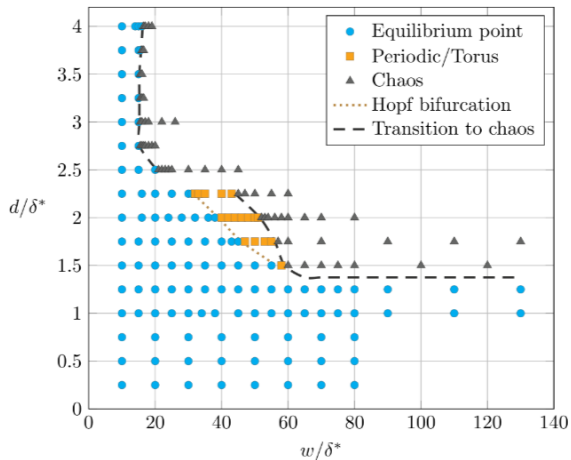


IMPERIAL

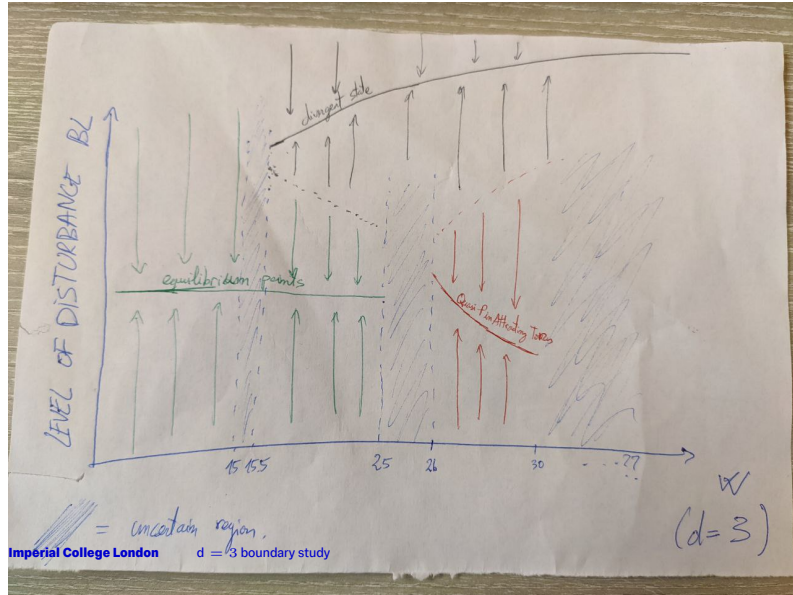
$d = 3$ **boundary study**

Reminder

- Lines $d = 2$, $d = 1.25$ look correct.

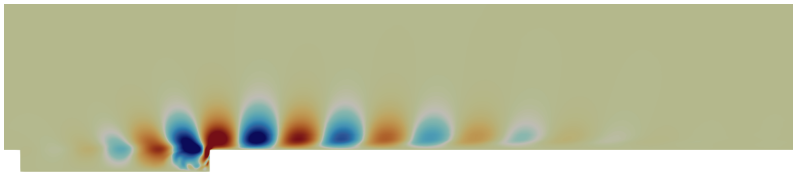


Sketch of the bifurcation diagram with the info so far



- Bif $w = 15.11 - 15.12$:
loss of globally attracting state?
- Bif $w = 25.41 - 25.42$:
It is actually a Hopf and what I was observing before was the real frequency coupled with a numeric frequency (sampling frequency). There is only one global unstable mode after $w = 25.42$.

$w = 26$ **global mode**



- DNS (blue)
- $Ae^{\sigma t} \cos(\omega t + \phi)$ (orange),
 $\sigma = 0.0011487$ (growth rate), $\omega = 0.263295$ (frequency).

